EXP 8

Consider a Online College Notice Portal. Develop a JavaScript program that will validate the controls in the forms you have created for the application. State the assumptions you make (business logic you are taking into consideration). Note: Your application must access a database using Servlet/JSP.

AIM:

To develop an Online College Notice Portal that validates user inputs using JavaScript on the client side and accesses a database using Servlet/JSP for storing and processing registration details.

ALGORITHM:

1. Start the application.
2. Display the registration form to the user.
3. Accept user inputs such as name, email, password, phone number, and exam details.
4. Perform client-side validation using JavaScript:
    - Check for empty fields.
    - Validate email format.
    - Validate password strength.
    - Validate phone number length and format.
5. If validation fails:
    - Display appropriate error messages.
    - Prevent form submission.
6. If validation is successful:
    - Submit the form to the server (Servlet/JSP).
7. On the server side:
    - Receive the form data.
    - Perform server-side validation.
    - Connect to the database.
8. Store user details in the database.
9. Generate confirmation message for successful registration.
10. Stop the application.

STEPS:

1. Design the registration form with fields like name, email, password, phone number, exam selection, and date.
2. Implement JavaScript validation for all form controls.
3. Ensure proper error messages are displayed for invalid inputs.
4. Configure Servlet/JSP to handle form submission.
5. Establish database connectivity using JDBC.
6. Create a database table for storing registration details.
7. Insert validated data into the database.
8. Retrieve and display confirmation details to the user.
9. Handle duplicate registrations and invalid data cases.
10. Test the application for different input scenarios.

ASSUMPTIONS (BUSINESS LOGIC)

1. Each user must register with a unique email ID.
2. Password must meet minimum security requirements (e.g., length, characters).
3. Phone number must be exactly 10 digits.
4. All fields are mandatory.
5. A user can register for only one exam at a time.
6. Exam date must be a valid future date.
7. Database stores all user and exam details securely.
8. Server-side validation is mandatory even after client-side validation.
9. Duplicate registrations are not allowed.
10. Proper error handling and user feedback are provided.

Program:

student.html:

```
<!DOCTYPE html>
<html>
<head>
<title>Student Registration</title>

<script>
```

```
function validateForm()
{
    var name=document.forms["f1"]["name"].value;
    var roll=document.forms["f1"]["roll"].value;
    var email=document.forms["f1"]["email"].value;
    var mobile=document.forms["f1"]["mobile"].value;
    var pass=document.forms["f1"]["pass"].value;
    var address=document.forms["f1"]["address"].value;

    var namePattern=/^[A-Za-z ]+$/;
    var emailPattern=/^[^ ]+@[^ ]+\.[a-z]{2,3}$/;
    var mobilePattern=/^[0-9]{10}$/;

    if(name=="" || !namePattern.test(name))
    {
        alert("Enter valid student name");
        return false;
    }

    if(roll=="")
    {
        alert("Enter Roll Number");
        return false;
    }

    if(!emailPattern.test(email))
    {
        alert("Enter valid email");
        return false;
    }

    if(!mobilePattern.test(mobile))
    {
        alert("Enter 10 digit mobile number");
        return false;
    }

    if(pass.length<6)
    {
        alert("Password minimum 6 characters");
```

```
        return false;
    }

    if(address=="")
    {
        alert("Enter address");
        return false;
    }

    return true;
}
</script>

<style>
body{
font-family:Arial;
background:#dbeafe;
text-align:center;
margin-top:50px;
}

.box{
background:white;
width:400px;
margin:auto;
padding:25px;
border-radius:10px;
box-shadow:0 0 10px gray;
}

input,select,textarea{
width:90%;
padding:8px;
margin:8px;
}
button{
padding:10px 20px;
background:#0d47a1;
color:white;
border:none;
```

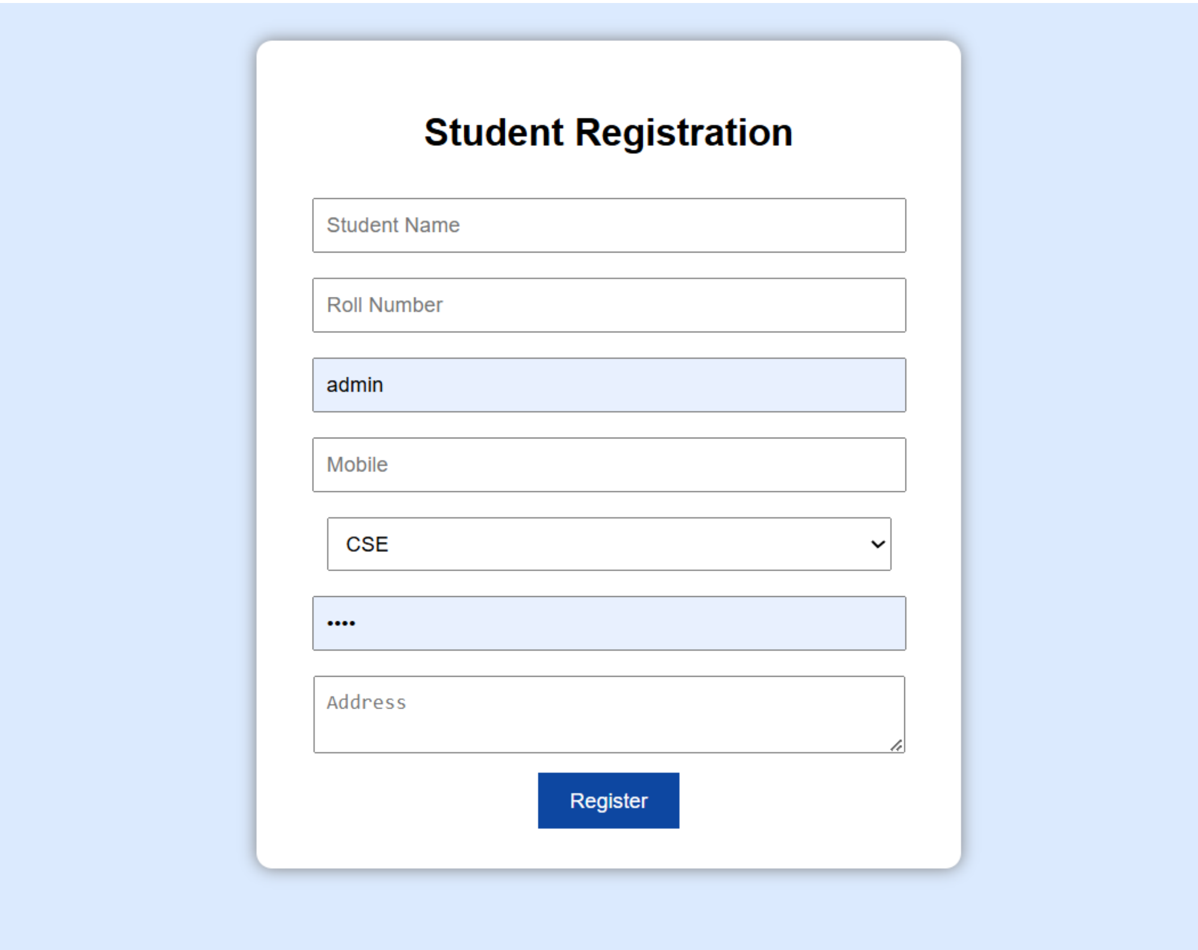
```
}
</style>

</head>
<body>

<div class="box">

<h2>Student Registration</h2>

<form name="f1" action="register" method="post" onsubmit="return validateForm()">

<input type="text" name="name" placeholder="Student Name">

<input type="text" name="roll" placeholder="Roll Number">

<input type="text" name="email" placeholder="Email">

<input type="text" name="mobile" placeholder="Mobile">

<select name="dept">
<option>CSE</option>
<option>IT</option>
<option>ECE</option>
</select>

<input type="password" name="pass" placeholder="Password">

<textarea name="address" placeholder="Address"></textarea>

<button type="submit">Register</button>

</form>

</div>

</body>
</html>
```

RegisterServlet.java:

```
import java.io.*;
import java.sql.*;
import jakarta.servlet.*;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.*;

@WebServlet("/register")
public class RegisterServlet extends HttpServlet {

protected void doPost(HttpServletRequest request,HttpServletResponse response)
throws ServletException,IOException
{
   response.setContentType("text/html");
   PrintWriter out=response.getWriter();

   String name=request.getParameter("name");
   String roll=request.getParameter("roll");
   String email=request.getParameter("email");
   String mobile=request.getParameter("mobile");
   String dept=request.getParameter("dept");
   String pass=request.getParameter("pass");
   String address=request.getParameter("address");

   try{
      Class.forName("com.mysql.cj.jdbc.Driver");

      Connection con=DriverManager.getConnection(
      "jdbc:mysql://localhost:3306/studentdb",
      "root",
      "");

      PreparedStatement ps=con.prepareStatement(
      "insert into student values(?,?,?,?,?,?,?)");

      ps.setString(1,name);
      ps.setString(2,roll);
      ps.setString(3,email);
      ps.setString(4,mobile);
```

```
        ps.setString(5,dept);
        ps.setString(6,pass);
        ps.setString(7,address);

        ps.executeUpdate();

        out.println("<h2>Student Registered Successfully</h2>");

        con.close();

    }catch(Exception e){
        out.println(e);
    }
}
}
```

OUTPUT:



All Notices

Title	Description	Date	Department
Exam Schedule Released	Final semester exams will begin from May 10.	2026-05-10	All Departments
Holiday Announcement	College will remain closed on May 1 for Labour Day.	2026-05-01	General
Workshop on AI	AI workshop conducted by industry experts on April 28.	2026-04-28	CSE
Sports Meet	Annual sports meet will be held from May 5 to May 7.	2026-05-05	Sports
Placement Drive	Top companies visiting campus next week for placements.	2026-05-12	Placement Cell
Library Notice	Library timing extended till 8 PM during exams.	2026-05-01	Library

RESULT

The Online Exam Registration Portal successfully validates user inputs using JavaScript and stores the validated data in a database using Servlet/JSP, ensuring accurate and secure registration.